Sidharth Nair

sidharthnair4306@gmail.com | Houston, TX | linkedin.com/in/sidharth-nair- | github.com/SNair-R

EDUCATION

The University of Houston, Houston, TX *Expected May 2028*

Bachelor of Science in Electrical Engineering — GPA: 3.93 / 4.00

- **Relevant Coursework:** Circuit Analysis I, Programming Applications in ECE, Computing for Engineers, Discrete Mathematics, Engineering Statistics

TECHNICAL SKILLS

- **Languages:** C, C++, Python, MATLAB
- **Embedded:** I2C, CAN, PWM, ADC, GPIO, Raspberry Pi Pico SDK (RP2040), Teensy 4.1, Jetson
- **Systems Concepts:** Memory management, non-blocking timing, function pointers, deterministic execution
- **Tools:** Git, GCC, Make, CMake, VS Code, Fusion 360

RESEARCH EXPERIENCE

Spark Up Research, Houston, TX — Undergraduate Researcher, Semiconductors & AI *Sep 2026 – Present*

- Developing an automated wafer-inspection system with a 3-person team spanning EE, mechanical and chemical engineering; building the AI defect-detection model that inspects wafers imaged under a microscope
- Training a computer-vision model to detect wafer defects on the Roboflow “Wafer Defect” die-level dataset, writing the image cropping and tiling pipeline in Python to match a Raspberry Pi camera and lens optical setup
- Configuring a robotic arm with a teammate to position wafers under the camera, and pursuing nanofab cleanroom training to validate the model on real wafers

PROJECTS

URC Rover Firmware Team | *C++, Python, CAN, Teensy 4.1, Jetson* *Aug 2026 – Present*

- Implemented fault handling on the rover’s Teensy 4.1 that separates C2 link loss from Jetson heartbeat loss as distinct fault states, keeping onboard autonomy running through the 2027 course’s planned C2 line-of-sight dropouts
- Added a Jetson heartbeat timeout and command-age checking over the rover’s 500 kbps CAN 2.0B bus, invalidating stale drive commands and moving the rover to a defined safe state, with the timeout documented as a tunable design value
- Verified invalid-packet rejection by injecting corrupted CAN frames and confirming the MCU discards them without updating the active drive command and reports a communication error
- Prioritized drive commands over 20 Hz telemetry on the CAN bus by managing a priority-ordered 11-bit message ID scheme, so safety-critical traffic wins arbitration

Pomodoro Study Timer | *C, Raspberry Pi Pico W, I2C, PWM, ADC, GPIO* *Aug 2026*

- Integrated a 16x2 I2C LCD, potentiometer input, debounced buttons, and a PWM buzzer into an interactive study timer
- Kept the display and buttons responsive through sessions up to 60 minutes by implementing non-blocking millisecond timing
- Signaled each phase of a 4-cycle study/break state machine with distinct buzzer patterns, so transitions are clear without looking at the screen

PC Hardware Monitor | *C, Python, Raspberry Pi Pico W, I2C* *Apr 2026*

- Built a desk display that shows live CPU/GPU temperature and usage on a 128x32 SSD1306 OLED, with on-screen warnings above 80°C (CPU) and 85°C (GPU)
- Delivered CPU/GPU statistics at 1-second intervals by writing a Python host script using WMI, psutil, and nvidia-smi and streaming comma-delimited packets over USB serial
- Rendered live readings over 100 kHz I2C by parsing the serial packets in C on the Pico and drawing them with a custom bitmap font

WORK EXPERIENCE

ACES Institute, Houston, TX — Cougar Tutor *Jan 2026 – May 2026*

- Tutored 5+ students in grades 3–6 daily in one-on-one and small-group settings, strengthening math fundamentals
- Diagnosed each student’s specific misunderstandings and adapted instruction, tracking progress across sessions

National Care EMS, Stafford, TX — EMT Basic *Jul 2025 – Oct 2025*

- Provided patient transport as an EMT across 40+ shifts, completing safety checks and equipment/supply verification before every transport under time pressure